Frank Shan

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Highlights

- Engineer focused on AI systems and automation, building retrieval, agentic workflows, and developer tooling that integrate LLMs with reliable backend infrastructure and measurable outcomes.

Education

University of Waterloo | *CGPA: 92.3/100* *Sep 2024 - Apr 2029*
Bachelor of Computer Science, Honours Computer Science, AI Specialization

Experience

Software Developer Intern | *Richmond, BC* *Sep 2025 – Apr 2026*
Ziyutec Inc.

- Built a low latency search system with hybrid lexical and vector retrieval and ranking, enabling automated resource discovery and recommendation workflows.
- Built a production-grade RAG system over 1,000+ enterprise documents, leveraging hybrid search, reranking, and context-aware chunking to optimize retrieval quality, improving faithfulness and contextual relevance by 16 percent (DeepEval).
- Engineered direct-to-object-storage uploads for large assets using AWS S3 presigned URLs; implemented end-to-end integrity validation (local and cloud checksum verification) to ensure data consistency at scale.
- Developed a workflow automation agent that orchestrates database queries and form submission via structured tool use, reducing manual operational workload by 20 percent across a 50-person team.

Research Assistant | *Waterloo, ON* *Feb 2026 – Present*
University of Waterloo

- Designed experiments for a probabilistic programming system (PAL) compiling symbolic constraints into tensor operations, enabling neuro-symbolic training with strict structural guarantees.
- Built a hybrid NER pipeline combining frozen language model embeddings with constraint-based decoding, improving prediction validity and early-stage training stability.

Projects

Mini VLM | *Python, PyTorch* *github*

- Built a LLaVA-style Vision-Language Model, integrating SigLIP-2 with Qwen3-0.6B via a modality projector, achieving 32.9 percent zero-shot MMStar accuracy under a 1B parameter constraint.
- Optimized training stability and memory usage via gradient checkpointing, mixed precision, and custom dataset pipelines, enabling efficient training on consumer GPUs.

Mini VLLM | *Python, PyTorch* *github*

- Built a lightweight LLM inference engine with continuous batching, request scheduling, and paged KV-cache architecture for efficient memory management.
- Implemented a custom scheduler and block manager to coordinate sequence execution and KV memory allocation, improving GPU utilization under concurrent workloads.

Snapplot | *React Native, Expo, TypeScript* *github*

- Built a real-time multiplayer system using WebSockets (Socket.io) to synchronize game state, player actions, and timed reveal events across clients with low-latency updates.

Skills

Programming Languages: Python, TypeScript, JavaScript, C++, SQL, HTML, CSS
Frameworks: NestJS, Next.js, React, Express, FastAPI, LangGraph, PyTorch
Tools: AWS, PostgreSQL, Linux, Docker, Git

Professional Certificates

Stanford Online *December 2025*
XCS224N Natural Language Understanding credential